CS 5594: BLOCKCHAIN TECHNOLOGIES

Spring 2021

THANG HOANG, PhD

COURSE OVERVIEW AND ORGANIZATION

Outline

- About Instructor
- High-level Objectives
- Grading
- (Tentative) Schedule
- Course Details
- Q&A

About Instructor

- **Assistant Professor**, CS Dept, Virginia Tech (Dec 2020 – current)
  - Applied Crypto Lab: We are recruiting Ph.D. and M.S. students!
  - Research Topics: Privacy-Enhancing Technologies, Applied Cryptography, Network Security
  - Publications, patents, open-source frameworks
- **Ph.D. (2020)**
  - University of South Florida (2019-2020)
  - Oregon State University (2015-2018)
  - Privacy-Preserving Functional Information Systems
- **M.S. (2014)**
  - Chonnam National University (S. Korea) (2012-2014)
  - Mobile Authentication with Machine Learning and Biometric Cryptosystem

Learning Objectives

- Understand principles of emerging blockchain technologies
- Harness blockchain on various applications domains (economics, healthcare)
- Design your own blockchains for specific application requirements

FOUNDATIONAL PRIMITIVES

- Distributed Systems
  - Peer-to-peer networks
  - Consensus
  - Security & Threat
- Cryptography
  - Hash function
  - Signatures

CORE TECHNIQUES

- Public blockchains
  - Architecture
  - How it works
- Private blockchains
  - Access control, consensus
- Smart contracts
  - Blockchain applications

ADVANCED TOPICS

- Confidential transactions
- Decentralized storage

Grading

- **NO** midterm and Final, but…
- **Homework (30%):** Tentative 3-4 HW problem sets with programming involved (Python or Java)
  - Ask to explore deeper in topics covered throughout the class
- **Presentation (20%):** Present paper(s) from top-tier security/blockchain conferences
  - Important chance to practice for future career!
- **Survey Paper (a team of two) (45%):** Extra-credit for research-oriented papers
  - Select a topic and write a detailed survey paper (6 pages IEEE double column style)
  - Develop a knowledge based on an important topic -> Practice executive reports
- **Participation (5%):** Constructive feedback for student presentations will be collected plus in-class engagement
- **Grading Scale:** A (90+); B(81-89); C(71-80); D(61-70); E (51-60); F(50-)

Course Topics (Tentative)

- **Week 1: Introduction**
  - History of Blockchain. What is Blockchain?
  - Why Blockchain?
- **Week 2-4: Fundamental Data Structures and Cryptographic Primitives**
  - Distributed systems, distributed consensus
  - Cryptographic hash, hash-based primitives
  - Public key cryptography, digital signatures
  - Elliptic Curve cryptography
- **Week 5-8: Blockchain Technologies**
  - Bitcoin as public blockchain basics
    - Network, address, transactions, blocks, consensus, mining, challenges
  - Other consensus protocols (PoUW, PoS, PoA)
  - Private blockchain

Course Topics (Tentative)

- **Week 5-8: Blockchain Technologies (cont.)**
  - Building decentralized/distributed applications with blockchain
    - Smart contracts
    - Ethereum, solidity
- **Week 9-12: Advanced Topics in Blockchain**
  - Confidential transactions
    - Anonymity and deanonymization
    - Tor, Silkroad
    - Privacy-preserving computation
    - Zero-knowledge proofs
  - Privacy-preserving blockchain platforms (Zcash, Monero, Hawk)
  - Decentralized storage and applications
- **Week 13-16: Student Presentation**

Student Presentation

- Finalize your presentation schedule soon
  - Volunteering preferred, or other policies will be implemented
- **No re-scheduling:** Only possible with a doctor note
- Select papers from top-tier cyber-security or blockchain conferences
  - Recommended List: ACM CCS, IEEE S&P, NDSS, USENIX Security, IEEE ICBC, IEEE BLOCKCHAIN, EuroS&P, Crypto, Eurocrypt, IEEE INFOCOM, ACSAC, IEEE ICDSC, PoPETs, Asiacrypt
  - Published between 2015 – 2020
  - Blockchain-related

Final Project

- Select some papers in the previous list, but publication date can be older
- Potential topic list (but not limited to)
  - Security & privacy of blockchain technology,
  - Blockchain platforms
  - Blockchain in specific domain (e.g., IoT, cryptocurrency, cloud computing, social networking, machine learning, etc)
  - Smart contracts

- Form a group of two, and inform the instructor your topic ASAP
  - Single-person project possible
  - Email your group info including **students' name and ID**, and your **selected topic** to thanghoang@vt.edu
  - Deadline: **Jan 29, 2021 (Fri) 11:59 PM EST**

Final Project

- **Theoretical analysis and comparison of existing results** ✓
- **Implementation and comparison of existing methods** ✓✓
  - Survey paper publications at the end of the course
- **New algorithm design, new system design** ✓✓✓
  - Publications at top-tier security or blockchain venues
- Different topic OK, but must be blockchain-related and allowed from your advisor
  - Your advisor may want to keep it secret (confidentiality requirement of your funding)
  - Do NOT bring it up unless you are permitted, or there will be in trouble for all of us!
- There will be interim report at the middle of semester, and you will be given one-to-one feedback on your report
  - Will be graded, so do NOT put off your writing

Final Project

- A good opportunity to master your research and writing skills (very important)
- A good guideline to research writing
  - https://www.darpa.mil/work-with-us/heilmeier-catechism

- **The Heilmeier Catechism**
  - What are you trying to do? Articulate your objectives using absolutely no jargon.
  - How is it done today, and what are the limits of current practice?
  - What's new in your approach and why do you think it will be successful?
  - Who cares? If you're successful, what difference will it make?
  - What are the risks and the payoffs?
  - How much will it cost? How long will it take?
  - What are the midterm and final "exams" to check for success?

Logistics and Notes

- Teaching Tools and Resources
  - Canvas
  - Course webpage: https://thanghoang.github.io/teaching/Spring21/CS5594/
- Teaching Team
  - Instructor: Thang Hoang, Ph.D.
    - Office: Suite 2202, VT Knowledge Works II Building
    - Email: thanghoang@vt.edu
    - Webpage: https://thanghoang.github.io
  - TA: Waad Aldndni
    - Email: waada@vt.edu
- Announcement & communication: via Canvas

Logistics and Notes

- Lecturer (Thang Hoang)
  - Office hours: **Friday @ 2 PM – 4 PM**
  - Zoom link: https://virginiatech.zoom.us/j/84859897740?pwd=Nk53bCtEWnArTGtZdUQva1l5Q0oxQT09

- TA (Waad Aldndni)
  - Office hours: **Monday @ 3 PM – 5 PM**
  - Zoom link: https://virginiatech.zoom.us/skype/6641881060

Logistics and Notes

- **Check course webpage and Canvas regularly**
  - Slides, research papers, assignment will be put at course webpage and announced on Canvas
- Register for Canvas Announcement
- Free online blockchain resources:
  - Crypto books
    - Introduction to Modern Cryptography, by Jonathan Katz and Yehuda Lindell
  - Blockchain books:
    - “Bitcoin and Cryptocurrency Technologies”, by Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder
    - “Mastering Bitcoin”, by Andreas M.Antonopoulos
  - Google, IACR, arxiv, etc
- Read syllabus!

Question?

- Question?